Statistics Quick Notes for Data Analysts

Quick Revision Notes · by Mahendra Singh · CrackAnalytics

Central tendency & spread

- **Mean** average — distorted by outliers · **Median** middle — robust (salaries!) · **Mode** most frequent
- **Variance** avg squared deviation · **SD** its root (same units) — low SD = consistent
- **IQR** = Q3–Q1; outlier if outside Q1–1.5·IQR to Q3+1.5·IQR

Distributions

- Normal: 68–95–99.7 rule for ±1/2/3 SD
- Right-skewed (income): mean > median · Left-skewed: mean < median
- **CLT:** sample means become normal as n grows (~30+), whatever the population — the basis of t-tests & confidence intervals

Hypothesis testing

- H0 = no effect · H1 = effect exists · run test → p-value
- **p-value** = probability of results this extreme if H0 true; $p < 0.05$ → reject H0 (significant)
- **Type I** false positive (α, claiming a fake win) · **Type II** false negative (β, missing a real win); power = $1-\beta$
- Common tests: t-test (means), chi-square (categories), ANOVA (3+ groups), z-test for proportions (A/B)

Correlation & regression

- Correlation $r \in [-1, +1]$ measures co-movement; **correlation ≠ causation** (ice cream vs drowning — summer!)
- Causation needs controlled experiments (A/B tests) or careful causal methods
- Linear regression $y = b0 + b1x$ — R^2 = % of variance explained
- Logistic regression outputs 0–1 probability — a CLASSIFIER despite the name

A/B testing quick recipe

- Define metric & minimum detectable effect → compute sample size BEFORE starting
- Randomize users, run full business cycles, don't peek daily and stop at first significance
- Report: lift, p-value, confidence interval — and business impact in ■

Metrics analysts must know

- Precision = TP/(TP+FP) — trust of positive predictions · Recall = TP/(TP+FN) — coverage of real positives
- F1 = harmonic mean; use when classes are imbalanced (fraud, churn) — accuracy lies there